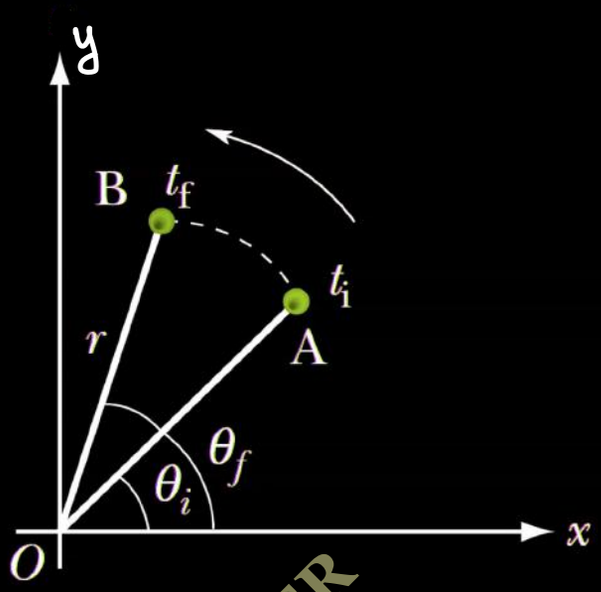
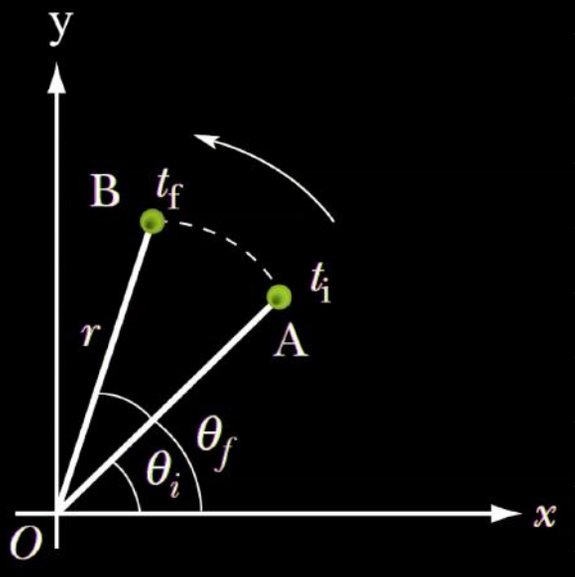
Angular Displacement (Scalar):



Suppose a particle is moving in a circle around the origin as shown in the figure. Let θ_i be the initial angular position of the particle as measured in anticlockwise sense from x-axis and let θ_f be the final angular position, then the angular displacement is simply $\Delta\theta = \theta_f - \theta_i$. Note that just by looking at picture of the particle at two distinct times is not sufficient to find angular displacement because advancing a particle by an angle $2n\pi$ would lead to exactly the same picture, thus one must be careful while using the term "angular position" in content of using it to calculate "angular displacement". An angle of $30°$ is not same as an angle of $390°$ in this content because the two would give a different value of angular displacement (and hence angular velocities).

Angular Velocity (Signed Scalar):



Suppose the angular position of a particle changes by an amount $\Delta\theta$ in a time interval Δt then we define its angular velocity as:

$$\omega_{avg} = \frac{\Delta\theta}{\Delta t} = \frac{\theta_f - \theta_i}{t_f - t_i}$$

Usually for circular motion in the plane of the page, anticlockwise angular displacements are considered as positive although, sometimes for convenience we might consider clockwise sense as positive (just as in projectile motion, sometimes we considered downward direction as +y axis).

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INSTANTANEOUS ANGULAR VELOCITY: Recall that instantaneous rate of change is also a kind of average rate of change but for an infinitesimal time interval. Thus we define instantaneous angular velocity as:

$$\omega = \frac{d\theta}{dt} = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t}$$

SI unit of angular velocity is radians per second.

ANGULAR ACCELERATION (SIGNED SCALAR): If the angular velocity of a particle changes by an amount $\Delta\omega$ in a time interval Δt then we define its average angular acceleration as $\alpha_{avg} = \frac{\Delta\omega}{\Delta t} = \frac{\omega_f - \omega_i}{t_f - t_i}$ we can also define instantaneous angular acceleration as average angular acc^n in an infinitesimal time interval thus:

$$\alpha = \frac{d\omega}{dt} = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t}$$

→ Now recall that $\omega = \frac{d\theta}{dt}$ thus we can also write $\alpha = d^2\theta/dt^2$

→ Also ∵ $\alpha = \frac{d\omega}{dt}$ we can write it as $\alpha = \frac{d\omega}{dt} \times \frac{d\theta}{d\theta} = \frac{d\theta}{dt} \cdot \frac{d\omega}{d\theta}$

thus $\alpha = \omega \frac{d\omega}{d\theta}$

→ Unit of α is rad/s^2.

ANALOGOUS DIFFERENTIAL EQUATION OF 1D MOTION AND CIRCULAR MOTION:

If we observe carefully, the differential equations governing kinematics of circular motion are exactly similar to the ones governing motion in 1D (thus their solutions and solution methods are also similar). Here we list out the analogues:

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① $v = \frac{ds}{dt}$	$\equiv$	$\omega = \frac{d\theta}{dt}$
② $a = dv/dt$	$\equiv$	$\alpha = d\omega/dt$
③ $a = v\,dv/ds$	$\equiv$	$\alpha = \omega\, d\omega/d\theta$
④ $a = d^2x/dt^2$	$\equiv$	$\alpha = d^2\theta/dt^2$

$x \equiv \theta$
$v \equiv \omega$
$a \equiv \alpha$
$s \equiv \Delta\theta$

KINEMATICS WITH CONSTANT ANGULAR ACCELERATION:

Since differential equations governing angular variables of circular motion are exactly similar to those governing linear variables in 1D motion, the solutions to the differential equations can also be written by one to one analogy. These analogues are listed as follows:

① $v = u + at$	$\equiv$	$\omega = \omega_0 + \alpha t$ (ω_0: initial angular velocity)
② $x = x_0 + ut + \frac{1}{2}at^2$	$\equiv$	$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$
③ $s = ut + \frac{1}{2}at^2$	$\equiv$	$\Delta\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
④ $v^2 - u^2 = 2as$	$\equiv$	$\omega^2 - \omega_0^2 = 2\alpha\Delta\theta$
⑤ $v_{avg} = \frac{u+v}{2}$	$\equiv$	$\omega_{avg} = \frac{\omega+\omega_0}{2}$
⑥ $s_{nth} = u + \frac{1}{2}a(2n-1)$	$\equiv$	$\Delta\theta_{nth} = \omega_0 + \frac{1}{2}\alpha(2n-1)$

RELATION BETWEEN SPEED AND ANGULAR VELOCITY:

Recall that if an arc subtends $\Delta\theta$ angle at centre of the circle then arc length is $\Delta l = R\Delta\theta$ where R is radius of the circle. Now if $d\theta$ is angular displacement in a time dt then the arc length covered is $|R\,d\theta| = dl$, thus instantaneous

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speed of the particle is simply $v = \frac{dl}{dt} = R\left|\frac{d\theta}{dt}\right|$

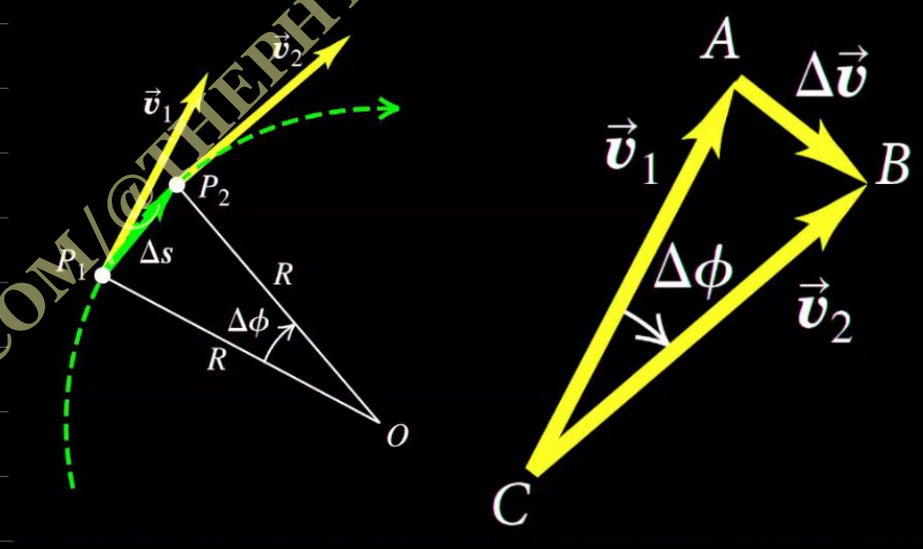
or $\boxed{v = |\omega| R}$ Now if total path length covered by a particle is to be found we can simply calculate it as

$$l = R\int |\omega| dt$$

Uniform Circular Motion: If a particle moves in a circle with constant speed (without sense reversal), the motion is called uniform circular motion (note that uniform circular motion is not an example of uniform motion because the direction of velocity is changing at every instant).

Velocity And Acceleration In Uniform Circular Motion:

Consider a particle moving with a constant speed v along arc of a circle covering a small angle $\Delta\phi$ in a time Δt. Note that velocity vector is always tangential to the path thus angle between velocity vectors at two times is exactly equal to the angle between radius vectors at the two times. In the figure above the right diagram is drawn in velocity space and it must be similar to the triangle OP_1P_2 (i.e $\Delta ABC \sim \Delta P_1P_2O$). Thus $\frac{|\Delta\vec{v}|}{v} = \frac{\Delta s}{R} = \frac{R\Delta\phi}{R} = \Delta\phi$



thus $\Delta\vec{v} = v\Delta\phi(-\hat{r})$ where $\hat{r}$ represents radially outward direction (Note that $\Delta\phi$ is infinitesimal and $\Delta\vec{v} \perp \vec{v}$ $\because \angle CAB \approx 90^\circ$) thus change in velocity vector is in the inward direction.

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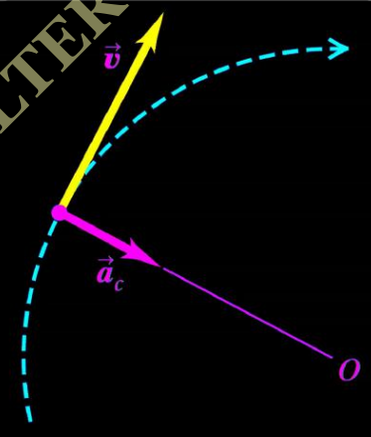
Now acceleration: $\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = v\frac{\Delta \phi}{\Delta t}(-\hat{r}) = v\omega(-\hat{r})$

Thus $\boxed{\vec{a} = v\omega(-\hat{r})}$

→ This acceleration is called centripetal acceleration ($\vec{a}_c$) because it is directed towards centre of the circle.

→ Also $v = ds/dt = R\,d\phi/dt = R\omega$ thus $v\omega = \omega^2 R$ this gives us two more ways of writing centripetal acceleration. $a_c = \omega^2 R = v^2/R$ thus:

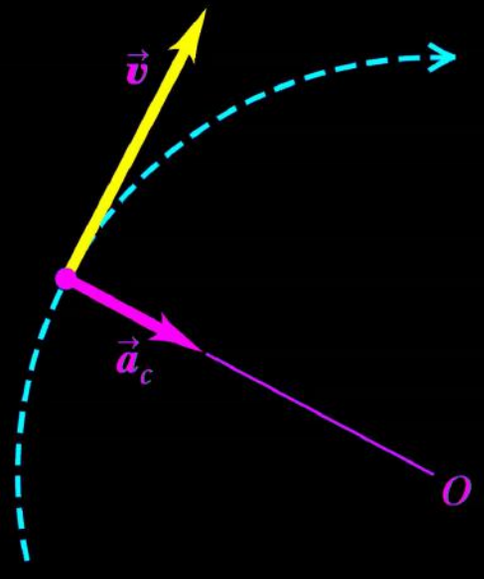
$$\boxed{\vec{a}_c = \omega v(-\hat{r}) = \omega^2 R(-\hat{r}) = \frac{v^2}{R}(-\hat{r})}$$



Pseudo Force In Rotating Frame In Uniform Circular Motion (Particle At Rest):

Imagine a person sitting at the centre of the circle on a revolving chair, the chair revolves such that the particle is always in front of the observer's eye. Thus in such a frame, the particle appears to be at rest. Now let $\vec{F}$ be the real force on the particle and let $\vec{F}_p$ be the required pseudo force on the particle then:

$$\vec{F} = m\omega^2 R(-\hat{r}) \quad (1) \quad (\because \vec{F} = m\vec{a})$$



Also:

$$\vec{F} + \vec{F}_p = 0 \quad (2) \quad [\because \vec{a} = 0 \text{ from rotating frame}]$$

(2) − (1) Gives:

$$\boxed{\vec{F}_p = m\omega^2 R\,\hat{r}}$$

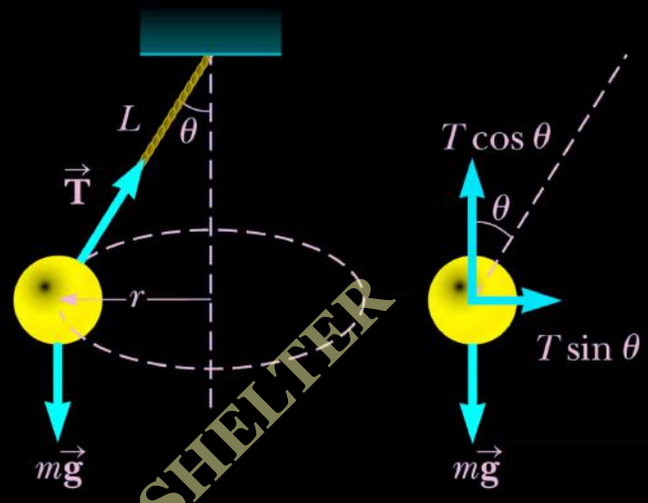
→ Also $\because v = \omega R$, $\vec{F}_p = mv\omega\hat{r} = \frac{mv^2}{R}\hat{r}$

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This force is called centrifugal force (since it is acting away from the centre of the circle).

Conical Pendulum:

A bob undergoes a circular motion in a "conical pendulum" arrangement as shown in the figure. The radius of the circle is r and speed of the particle is v. We wish to find angle of the string with vertical. We could either do it from ground frame, or approach it from rotating frame. Consider the FBD from ground frame first. Vertical force balance gives: $T\cos\theta = mg$ ① also from horizontal acceleration: $T\sin\theta = \frac{mv^2}{r}$ ②

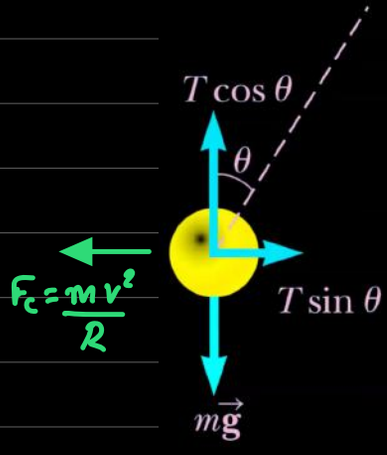


From ① & ②, $\tan\theta = v^2/rg$ ans.

Drawing FBD from rotating frame, radial force balance equation gives:

$$T\sin\theta = F_c = \frac{mv^2}{R} \quad ③$$

Thus equation ③ is same as ② though obtained using logic of equilibrium.



Minimum Angular Velocity For Conical Pendulum:

We got $\tan\theta = \frac{v^2}{rg}$ and $v = \omega r$ thus:

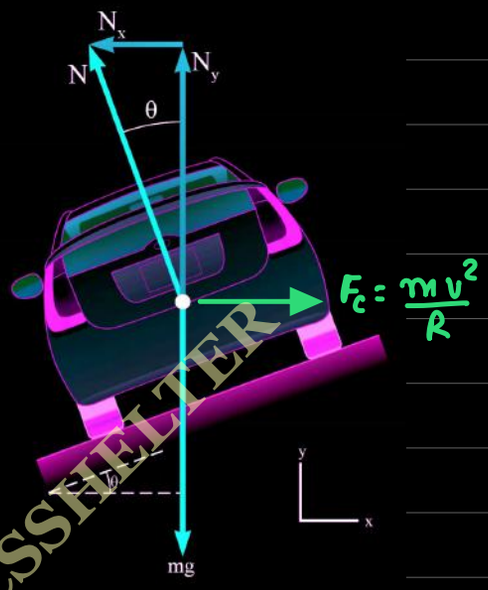
$$\frac{\sin\theta}{\cos\theta} = \frac{\omega^2 L\sin\theta}{g} \Rightarrow \cos\theta = \frac{g}{\omega^2 L} < 1 \quad (\because \cos\theta < 1)$$

thus $\omega > \sqrt{g/L}$, this gives the minimum possible angular speed and hence maximum possible time period of conical pendulum, $T = 2\pi/\omega$

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thus $T_{mean} = 2\pi\sqrt{L/g}$ which is also the time period of a usual pendulum. Thus a conical pendulum has generally a faster cycle than a simple pendulum.

BANKING OF ROADS: In hilly areas, the roads are usually banked (tilted with horizontal) so as to help vehicles safely execute the turns even when the roads are slippery. We wish to find banking angle θ of the road such that the car can safely execute a turn of radius R without bringing friction in to play. We consider the FBD of the car from a rotating frame at centre of curvature, such that car appears to be at rest from this frame. Thus balancing horizontal force:



$$F_c = \frac{mv^2}{R} = N\sin\theta \quad (1)$$

Also $mg = N\cos\theta$ (2) Now (1)/(2) gives:

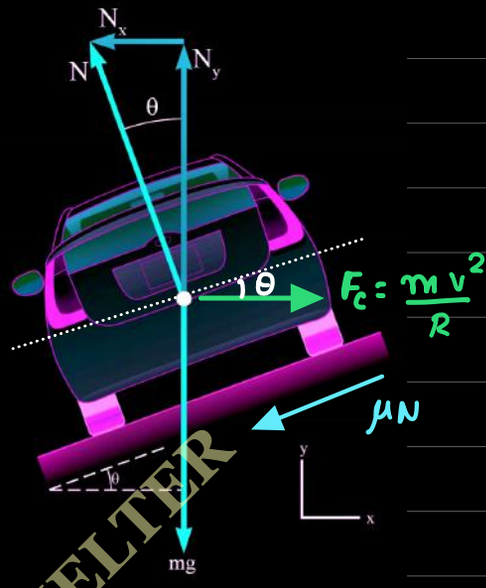
$$\boxed{\tan\theta = v^2/Rg}$$

→ This is the required banking angle for safe moving of a car with velocity v without calling friction in to play. With friction coming in to action, we will see that for any banking angle, there will be a range of possible speeds on the banked road.

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Velocity Bounds With Friction On Banked Roads:

Consider a car moving in a radius R. Centrifugal force varies with square of the speed thus if speed is large enough, component of centrifugal force along the incline will cause an outward skidding tendency thus at the verge of skidding, the limiting friction will act in downward direction. Now balancing forces along the incline:



$$mg\sin\theta + \mu N = \frac{mv^2}{R}\cos\theta \quad (1)$$

also

$$N = mg\cos\theta + \frac{mv^2}{R}\sin\theta \quad (2) \quad \text{(Balancing } \perp \text{ to incline).}$$

From (1) & (2):

$$\boxed{v_{max} = \sqrt{gR}\sqrt{\frac{\mu + \tan\theta}{1 - \mu\tan\theta}}} \quad (A)$$

→ Now recall that $\mu = \tan\phi_s$ where ϕ_s is angle of repose and also $\tan(A+B) = (\tan A + \tan B)/(1 - \tan A \tan B)$ using this we can further simplify the above expression:

$$\boxed{v_{max} = \sqrt{gR\tan(\theta + \phi_s)}} \quad (B)$$

→ If $\theta + \phi_s = 90°$, we can see that v_{max} goes to infinity, which means there is no upper bound for velocity.

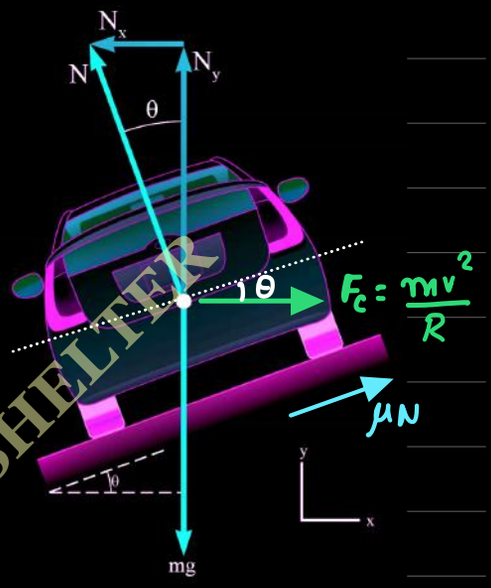
→ Now what if $\theta + \phi_s > 90°$, in that case imagine the plane to have a smaller coefficient of friction $\mu' = \tan\phi_s'$ such that $\theta + \phi_s' = 90°$. Which means infinite velocity would be possible even with smaller coefficient of friction, which also means that if $\theta + \phi_s > 90°$ then infinite velocity is possible without reaching the limiting friction.

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→ Equations (A) and (B) above can be used for finding out minimum required coefficient of friction to drive at certain maximum speed:

$$\phi_s = \tan^{-1}\left[\frac{v_{max}^2}{gR}\right] - \theta \quad \text{and} \quad \mu_{min} = \tan\phi_s.$$

Minimum Driving Speed:



If the speed of the vehicle is too small, the centrifugal force will become too small and if the component of centrifugal force along the incline becomes smaller than $mg\sin\theta$, the vehicle will develop a downward skidding tendency and friction will have to act up the plane. The FBD of the car in limiting case is shown in the figure. If we wish we could balance the forces along the incline and perpendicular to incline, however we will do it in a smarter manner. Recognize that the only difference in this FBD and the FBD for calculating v_{max} is the change in direction of μN. This FBD is as if μ were replaced by $-\mu$ in the previous FBD and the same effect will carry throughout the equations and their solution. Thus we can directly write the solution for v_{min} by the substitution $\mu \to -\mu$. Thus from (A):

$$v_{min} = \sqrt{gR\left[\frac{\tan\theta - \mu}{1 + \mu\tan\theta}\right]}$$

Now since $\mu = \tan\phi_s$ thus:

$$v_{min} = \sqrt{gR\tan(\theta - \phi_s)}$$

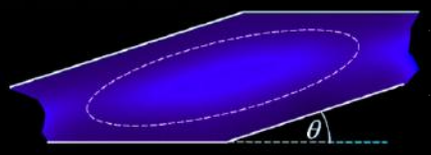
→ If $\theta = \phi_s$ the $v_{min} = 0$, this simply means that if banking angle is equal to angle of repose, it is just possible to leave

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a car parked. However if $\theta - \phi_s < 0$ then we can imagine the plane to have some lower coefficient of friction $\mu' = \tan\phi_s'$ such that $\theta - \phi_s' = 0$, which means it would have been possible to park the vehicle even with a lower coefficient of friction.

Example: Pathfinder: MCQ 36:

A large parking place has uniform slope of angle θ with the horizontal. A driver wishes to drive his car in a circle of radius R, at constant speed. Coefficient of static friction between the tyres and the ground is μ. What greatest speed can the driver achieve without slipping? Assume entire load of the car on the front wheels.



(a) $\sqrt{gR\tan\theta}$ (b) $\sqrt{gR\cot\theta}$

(c) $\sqrt{gR(\sin\theta+\mu\cos\theta)}$ (d) $\sqrt{gR(\mu\cos\theta-\sin\theta)}$

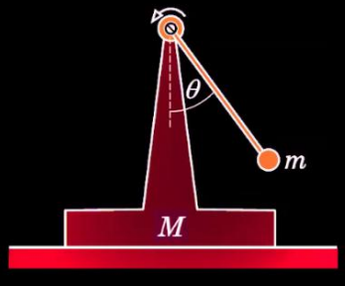
Solution: Here, recognize that resultant of centrifugal force and mg is to be balanced by frictional force which has a limiting value $\mu N = \mu mg\cos\theta$. Now resultant of centrifugal force and mg has largest magnitude when car is at the bottommost point (∵ both forces have constant magnitude and at bottommost point both are perfectly aligned). Now considering limiting equilibrium:

$$mg\sin\theta + \frac{mv^2}{R} = \mu mg\cos\theta \Rightarrow v = \sqrt{gR[\mu\cos\theta - \sin\theta]}$$ Ans

(option d) Ans.

Example: Pathfinder: Build: SS:

A motor is installed at the top of a pole rigidly fixed on a platform. A light rod of length $r = 1$ m is rigidly attached to the motor shaft at its one end and at the other end a small ball of mass m is attached. The rod can be rotated in a vertical plane with the help of the motor. Total mass of the platform, the pole and the motor is $\eta = 4$ times the mass of the ball. The motor rotates the rod at a constant angular velocity. The platform is placed on a horizontal surface, where coefficient of friction is $\mu = 1/\sqrt{3}$. At which minimum angular velocity ω_0 of the rod will the platform start sliding?



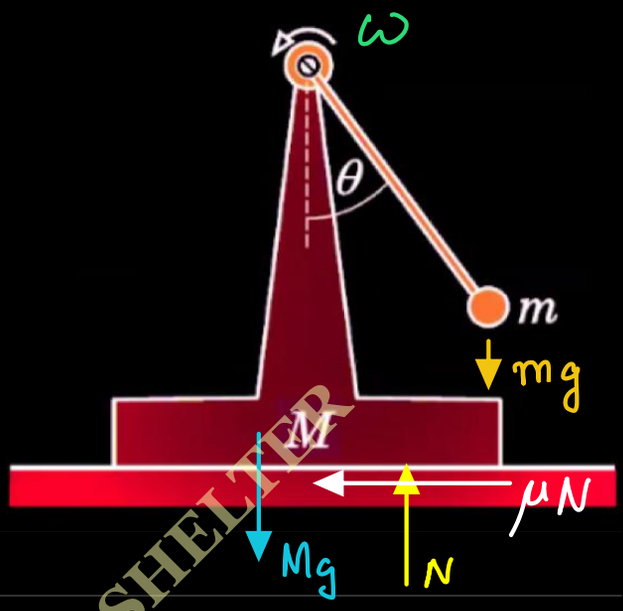
Solution: Here once again we will try to make some smart equations avoiding the internal unknown forces in the system.

We will try to find relation between ω and μ for

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impending slipping and find θ where maximum μ is required (or ω needs to be minimum for a given μ).

Now using $\Sigma F_{x_i} = \Sigma m_i a_{x_i}$ in limiting equilibrium:



$\mu N = m\omega^2 r \sin\theta$ ① Similarly in y direction:

$N - (M+m)g = m\omega^2 r \cos\theta$ ②

Now we can solve ① & ② for ω & N we get:

$$\omega^2 = \frac{\mu(M+m)g}{mr[\sin\theta - \mu\cos\theta]}$$

Now we need to minimize ω^2 wrt θ for no slipping at any θ. For that we need to maximize $\sin\theta - \mu\cos\theta$. Let $\sin\theta - \mu\cos\theta = q$

$\frac{dq}{d\theta} = 0 \Rightarrow \cos\theta + \mu\sin\theta = 0 \Rightarrow \tan\theta = \frac{-1}{\mu} \Rightarrow \theta \in (90°, 180°)$

Thus $\cos\theta = \frac{-\mu}{\sqrt{\mu^2+1}}$, $\sin\theta = \frac{1}{\sqrt{\mu^2+1}} \Rightarrow \sin\theta - \mu\cos\theta = \sqrt{\mu^2+1}$

Thus $\omega_{max} = \sqrt{\frac{\mu(M+m)g}{\sqrt{1+\mu^2}\, mr}}$ Putting $M = \eta m$, $\omega_{max} = \sqrt{\frac{\mu(1+\eta)g}{\sqrt{1+\mu^2}\, r}}$ Ans.

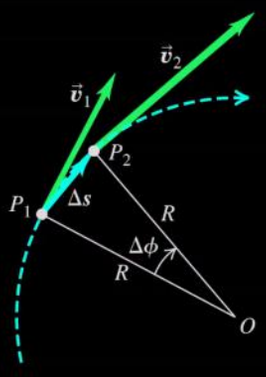
Putting the values, we get $\omega_{max} = 5$ rad/s!

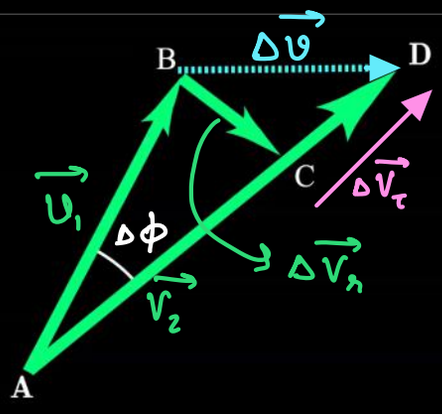
→ Note that we minimize ω wrt θ to get maximum permissible ω because, ω cannot exceed its allowable value at any θ).

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Tangential Acceleration:

Consider a particle moving in a circle whose speed changes from v to $v+\Delta v$ as it covers an arc of angle $\Delta\phi$. The diagram for change in its velocity vector is shown in the adjacent figure. Here





the change in velocity vector is given by $\vec{BD}$ which in turn can be written as $\vec{BC}+\vec{CD}$. Now we obtain $\vec{BC}$ by placing compass at A and forming an arc of radius AB. Thus for sufficiently small $\Delta\phi$, BC is $\perp$ to AB and hence towards centre of the circle, also $BC = AB\times\Delta\phi$ ($\because l = r\Delta\phi$) thus $BC = v\Delta\phi = AC\,\Delta\phi$, Also $AD = v+\Delta v$ thus $CD = \Delta v$ and since BC is radially inwards, CD must be along tangential direction, and have a magnitude Δv ($\because CD = AD - AC = v+\Delta v - v = \Delta v$). Thus change in velocity vector is:

$$\Delta\vec{v} = v\Delta\phi(-\hat{r}) + \Delta v\,\hat{\tau}$$ where $\hat{\tau}$ represents tangential direction. Thus now acceleration:

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t} = v\frac{\Delta\phi}{\Delta t}(-\hat{r}) + \frac{\Delta v}{\Delta t}\hat{\tau},$$ now in the limit $\Delta t\to 0$,

$$\boxed{\vec{a} = v\omega(-\hat{r}) + \frac{dv}{dt}\hat{\tau}},$$ Here we see that acceleration has two components, a centripetal component $v\omega$ and a tangential component dv/dt. Let us call them a_c and a_τ respectively.

Tangential Acceleration: Just now we derived tangential acceleration as $a_\tau = dv/dt$ here dv was change in (tangential) speed of the particle. Thus if particle is speeding up tangential acc^n is along the tangent of the path while if particle is slowing down, it is against the tangent.

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Blunder Alert: There can be some confusion here due to notation. Here we are using dv for change in speed while $d\vec{v}$ for change in velocity vector. Note that with this choice of notation $|d\vec{v}| \neq dv$. Thus always read dv/dt as $d(\text{speed})/dt$. Even when speed is constant $d\vec{v} \neq 0$ while $dv = 0$. Recall that velocity vector was changing even in uniform circular motion.

Relation Between Tangential Acceleration And Angular Acceleration:

We proved that $a_t = dv/dt$ however $v = R\omega$ thus, $dv/dt = R\, d\omega/dt$, and $d\omega/dt$ is nothing but angular acceleration thus we can also write $a_t = R\alpha$. Thus:

$$\boxed{a_t = dv/dt = R\alpha}$$

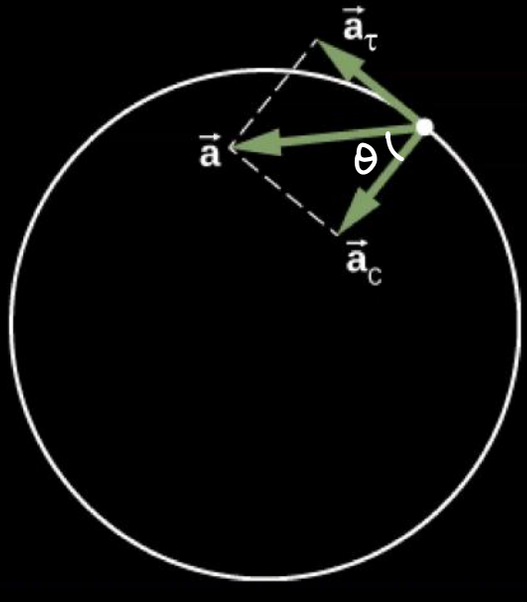
Centripetal Acceleration (Normal / Radial Acceleration):

We proved $a_c = v\omega$, also $v = \omega R$ thus we have alternate formulae for centripetal acceleration as:

$$\boxed{a_c = v\omega = \omega^2 R = v^2/R}$$

Total Acceleration: Total acceleration is simply the resultant of tangential and centripetal acceleration. Thus total acceleration magnitude:

$$\boxed{a = \sqrt{a_c^2 + a_t^2}}$$



and the angle between total acceleration and radial acceleration, $\boxed{\tan\theta = a_t/a_c}$

→ If angle between total acceleration and velocity vector is obtuse, particle is slowing down, if it is acute angle, particle is speeding up.

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EXAMPLE: A car was moving with a constant speed u in a circle of radius R. Now at $t=0$, it begins to speed up such that angle between total acceleration and centripetal acceleration is always θ. Find (a) speed as a function of time (b) Time at which the car begins to skid if coefficient of friction is μ and there is no banking.

Solution: $\frac{a_T}{a_c} = \tan\theta \Rightarrow \frac{dv}{dt} = \frac{v^2}{R}\tan\theta \Rightarrow \frac{dv}{v^2} = \frac{1}{R}\tan\theta\, dt$

$$\Rightarrow \int_u^v \frac{1}{v^2}dv = \frac{1}{R}\tan\theta \int_0^t dt \Rightarrow -\frac{1}{v} + \frac{1}{u} = \frac{t\tan\theta}{R}$$

Thus $v = \frac{uR}{R - ut\tan\theta}$ ① Ans.

Now for slipping, $a = \mu mg/m = \mu g$ thus, $a_c^2 + a_T^2 = \mu^2 g^2$

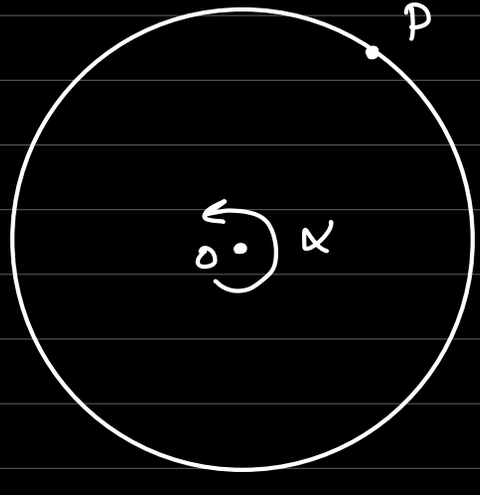
$\Rightarrow a_c^2(1 + \tan^2\theta) = \mu^2 g^2$ $(\because a_T = a_c \tan\theta) \Rightarrow a_c = \frac{\mu g}{\sec\theta}$

$\Rightarrow \frac{v^2}{R} = \mu g \cos\theta$ ② Solving ① & ②,

$$t = \frac{R}{u\tan\theta} + \sqrt{\frac{R}{\mu g \sin\theta \tan\theta}}$$ Ans.

EULER FORCE IN CIRCULAR MOTION:

Imagine a particle P moving with an angular acceleration α in a circle of radius R and an observer at O always keeping eye on P then from such a frame, the particle appears at rest. Now from ground frame, in tantential direction, $\vec{F}_T = m\vec{a}_t = mR\alpha\hat{\tau}$, and from the



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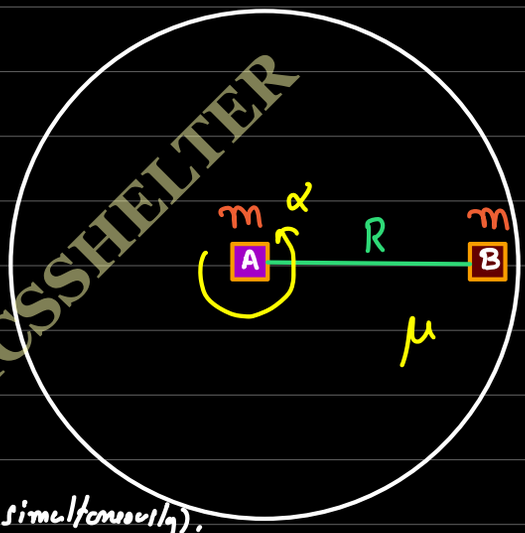
rotating frame, tangential acceleration appears to be zero, now let $\vec{F}_E$ be the required pseudo force in this frame then to satisfy Newton's second law:

$$\vec{F}_t + \vec{F}_E = 0 \Rightarrow \vec{F}_E = -\vec{F}_t$$

thus $\boxed{\vec{F}_E = -mR\alpha\,\hat{t}}$

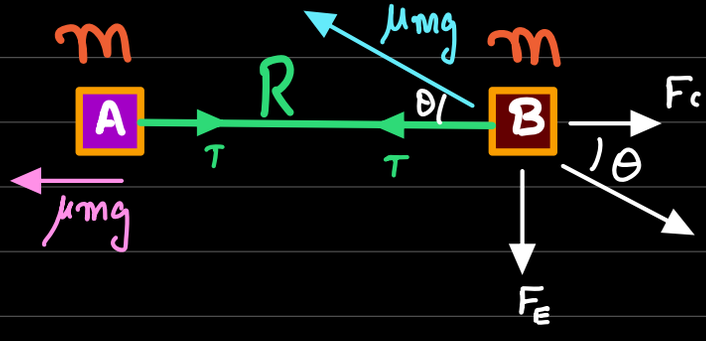
→ This pseudo force is called Euler force and acts against the direction of tangential acceleration.

ORIGINAL : A turn table with coefficient of friction μ has two blocks connected with a relaxed inextensible cord as shown in the figure. If the turn table starts from rest with angular acceleration α, after how much time does the slipping of a blocks start? [Assume that blocks slip simultaneously].



Solution: As the turn table rotates, the block B experiences an increasing centrifugal force and a constant tangential euler force in the rotating frame of the turn table.

Now let the direction of impending slipping of block B be at an angle θ from the radial direction.



Writing equilibrium equations at impending slip: $\Sigma \vec{F}_{in} = 0$

$$\Rightarrow F_c - \mu mg\cos\theta - \mu mg = 0 \quad (1)$$

Similarly in tangential direction:

$$F_E = \mu mg \sin\theta \quad (2)$$

Solving (1) & (2): $F_c = \mu mg\left[1 + \sqrt{1 - (F_E/\mu mg)^2}\right]$ (3)

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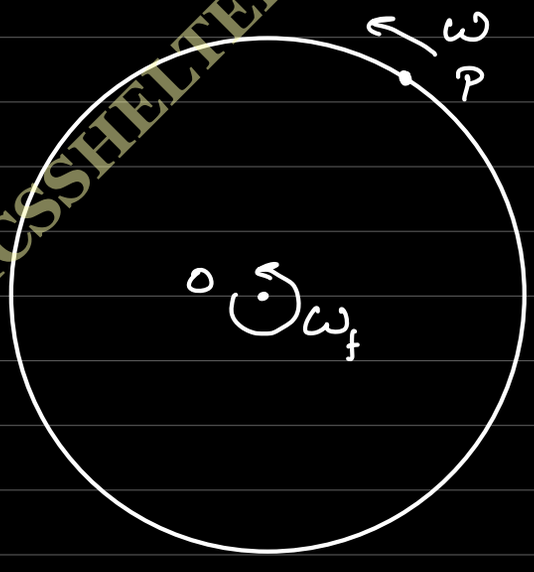
and $\theta = \tan^{-1}\left[\frac{1}{\sqrt{(\mu mg/F_e)^2 - 1}}\right]$ ④ (Rejected negative value of $\tan\theta$).

Now $F_c = m\omega^2 R = m(\alpha t)^2 R$ and $F_E = mR\alpha^2$

putting in ③ and solving for t: $t = \frac{\mu g - \sqrt{\mu^2 g^2 - R^2\alpha^2}}{R\alpha^2}$ Ans.

CORIOLIS FORCE IN CIRCULAR MOTION:

Consider a particle P rotating in a circle of radius R with an angular velocity ω. An observer located at O, is revolving with an angular velocity ω_f while observing the particle P. Now as seen from ground frame, acc^n of P is $\omega^2 R(-\hat{n})$ thus if $\vec{F}$ be the net force on the particle then:



$$\vec{F} = m\omega^2 R(-\hat{n}) \quad ①$$

From the revolving observer's frame, particle is moving with $\omega - \omega_f$ accordingly the apparent acceleration of the particle is $(\omega-\omega_f)^2 R(-\hat{n})$ Now $\vec{F}_p$ be the required pseudo force in this frame then:

$$\vec{F} + \vec{F}_p = m(\omega-\omega_f)^2 R(-\hat{n}) = m\omega^2 R(-\hat{n}) + m\omega_f^2 R(-\hat{n}) - 2m\omega\omega_f R(-\hat{n}) \quad ②$$

Subtracting: ② − ①:

$$\vec{F}_p = m\omega_f^2 R(-\hat{n}) - 2m\omega\omega_f R(-\hat{n})$$

$$\Rightarrow \vec{F}_p = m\omega_f^2 R\hat{n} - 2m\omega_f^2 R(\hat{n}) + 2m\omega\omega_f R\hat{n}$$

$$\Rightarrow \vec{F}_p = \vec{F}_c - 2m\omega_f(\omega-\omega_f)R\hat{n}$$ where $\vec{F}_c$ is centrifugal force.

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The second term in the above expression is called coriolis force. This can be written as:

$\vec{F}_{cor} = -2m\omega_f v_{rot}\hat{r}$ since $(\omega - \omega_f)R$ is velocity as seen from rotating frame. Vectorially it can be readily verified that coriolis force is simply $-2m\vec{\omega}_f \times \vec{v}_{rot}$. Thus:

$$\boxed{\vec{F}_{cor} = -2m\vec{\omega}_f \times \vec{v}_{rot}}$$



→ Recall from our discussion on 2D kinematics in polar coordinates, $\vec{\omega}$ and $\vec{\alpha}$ were defined using right hand thumb rule i.e curl the fingers of right hand in sense of ω and thumb gives the direction of $\vec{\omega}$ (similarly the relation between α and $\vec{\alpha}$).

→ Recall from discussion of 2D kinematics in polar coordinates that acceleration in general plane motion was given as:

$\vec{a} = (\ddot{r} - \omega^2 r)\hat{r} + \vec{\alpha} \times \vec{r} + 2\vec{\omega} \times \vec{v}_{rot}$, circular motion is a special case of that. Here $\ddot{r} = 0$ and $\vec{v}_{rot} = 0$ ($\because r$ = constant)

Thus:

$\vec{a} = -\omega^2 r\hat{r} + \vec{\alpha} \times \vec{r}$ (or $r\alpha\hat{\theta}$).

→ Recall from discussion of pseudo forces in rotating frames:

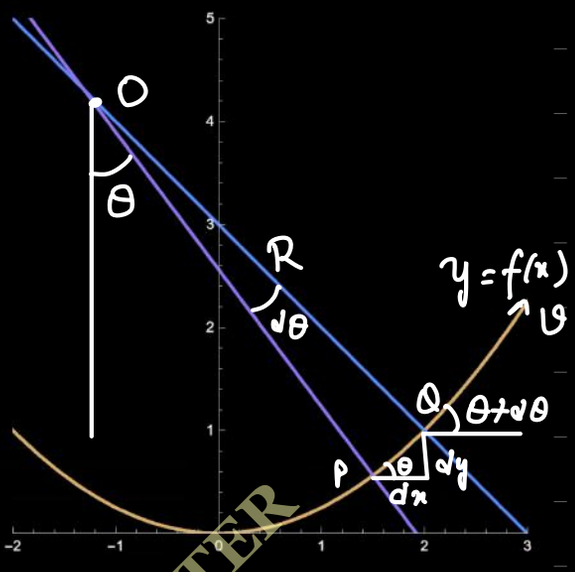
$\vec{F}_p = -m\vec{a}_0 - m\vec{\omega} \times (\vec{\omega} \times \vec{r}) - m(\vec{\alpha} \times \vec{r}) - 2m\vec{\omega} \times \vec{v}_{rot}$

the pseudo forces obtained in circular motion are special cases for this formula where $a_0 = 0$. The other three terms are centrifugal, euler and coriolis forces.

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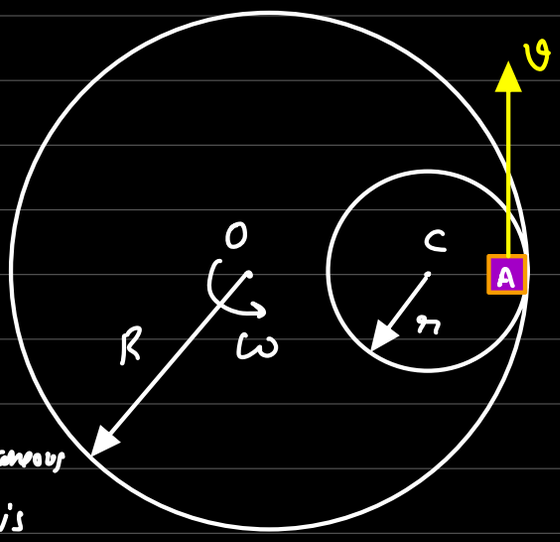
Radius Of Curvature Of Arbitrary Curve:

Recall from 2D kinematics that any arbitrary curve can be thought of as made up infinite number of infinitesimal circular arcs with radius of curvature being defined by intersection point of two normals drawn at arbitrarily close points. Thus formulae derived for circular motion also apply to infinitesimal segments of arbitrary path. Thus if a particle is moving along an arbitrary path with some speed v and has a centripetal acceleration a_c then its radius of curvature is simply $\boxed{R = v^2/a_c}$.

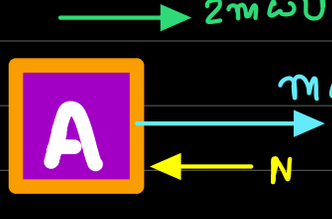


→ Recall that radius of curvature of arbitrary curve is $\boxed{R = \frac{(1+y'^2)^{3/2}}{y''}}$

Example: A turn table of radius R is rotating with constant angular velocity ω while a block is executing circular motion from the frame of the turn table in a smooth circular groove under action of normal reaction, centrifugal force and coriolis force only. If at farthest point from O, the instantaneous speed of the block from rotating frame is v, find (a) Normal reaction from side wall of the groove (b) instantaneous radius of curvature of the block's trajectory from ground frame.



Solution: From rotating frame of the turn table, the horizontal forces on the block are shown in adjacent figure.



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From this frame, block has an inward acceleration mv^2/r. Thus:

$$N - m\omega^2 R - 2m\omega v = mv^2/r \Rightarrow N = m\left[\frac{v^2}{r} + \omega^2 R + 2\omega v\right]$$ Ans.

Now from ground frame, N is the only net force on the block in horizontal plane acting perpendicular to its velocity thus centripetal acceleration from ground frame is simply N/m. also speed of the block from ground frame is $v' = v + \omega R$, thus required radius of curvature from ground frame:

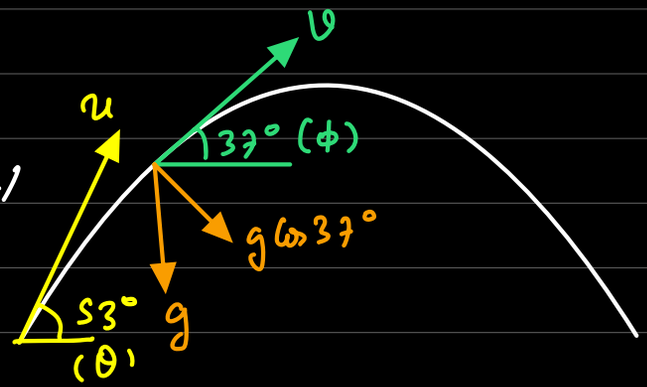
$$R_c = v'^2/(N/m) = \frac{(v+\omega R)^2}{v^2/r + \omega^2 R + 2\omega v}$$ Ans.

EXAMPLE: A projectile is thrown at an angle $\theta = 53°$ with a speed $u = 10\,ms^{-1}$. What is radius of curvature of its trajectory when it is flying at an angle $\phi = 37°$ above horizontal.

Solution: Since horizontal speed of a projectile is constant, $u\cos\theta = v\cos\phi$

$\Rightarrow v = u\cos\theta/\cos\phi$. Now centripetal acceleration is $g\cos\phi$ thus Radius of curvature is:

$$R = v^2/a_c = \frac{u^2\cos^2\theta}{\cos^2\phi \times g\cos\phi}$$



$$\Rightarrow \boxed{R = \frac{u^2\cos^2\theta}{g\cos^3\phi}} = 100 \times \frac{9}{25} \times \frac{125}{64} \times \frac{1}{10} = 225/32\ m$$ Ans.

$\approx 7.03\ m$ Ans.

EXAMPLE: A projectile is fired at an angle θ to have a range R (ground to ground), what is its radius of curvature when its horizontal displacement is ηR?

Solution: $R = (1+y'^2)^{3/2}/y''$, $y = x\tan\theta\,(1 - x/R)$

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Thus $y = x\tan\theta - \frac{x^2\tan\theta}{R} \Rightarrow y' = \tan\theta - \frac{2x\tan\theta}{R}$

for $x = \eta R$, $y' = \tan\theta\,(1-2\eta)$. $y'' = -\frac{2\tan\theta}{R}$

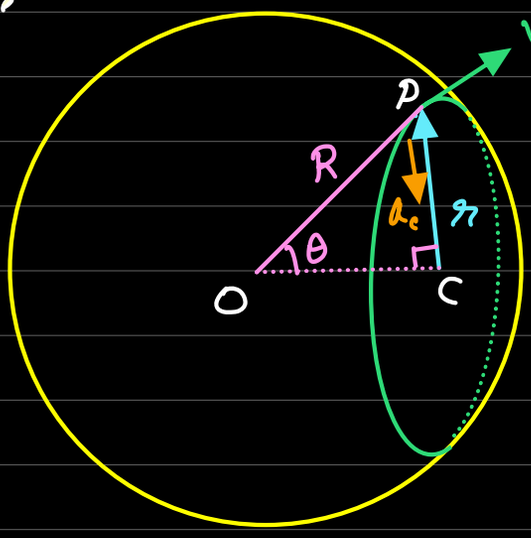
Thus required radius of curvature:

$$R_c = -\frac{R\left[1+\tan^2\theta\,(1-2\eta)^2\right]^{3/2}}{2\tan\theta}$$ Ans.

It has maximum magnitude for $\eta = 1$ or 0 and minimum for $\eta = 1/2$. (Radius of curvature is min at highest point and max at launch and landing points).

Central Acceleration Of A Point Moving On Surface Of A Sphere:

If a particle is moving on surface of a sphere, its elemental path can be imagined to be part of a circle drawn on the (see the green circle). Now it will have a centripetal acceleration $a_c = v^2/r$ towards C along with some tangential acceleration. To find the net component of acceleration towards O, we need to take components of these accelerations along $\vec{PO}$. Now the tangential direction is $\perp$ to $\vec{PO}$ thus tangential acceleration has no component along $\vec{PO}$ while centripetal acceleration has a component $a_c \sin\theta$, and $\sin\theta = r/R$. Thus acceleration component of particle towards O is simply:



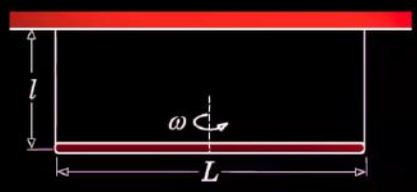
$$\vec{a}_0 = \frac{v^2}{r}\times\frac{r}{R}(-\hat{R}) \quad \text{i.e} \quad \frac{v^2}{R}(-\hat{R})$$. Thus whenever any particle is moving along surface of a sphere, its component of acceleration towards centre of the sphere is $\boxed{\vec{a}_0 = \frac{v^2}{R}(-\hat{R})}$

→ Here R is the radius of the sphere and $-\hat{R}$ is the radial unit vector (towards centre of the sphere).

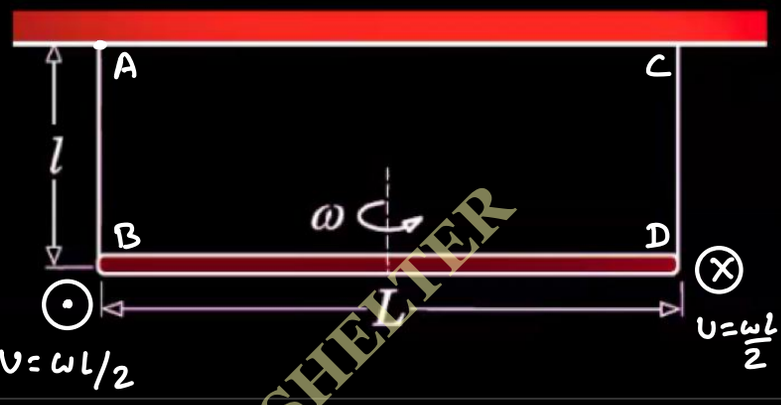
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Example: Pathfinder : Check : 22 :

A rod of mass m and length L is suspended from the ceiling with the help of two light inextensible cords each of length l so that the rod is horizontal. The rod is given an angular velocity ω about its central vertical axis. Find increment in the tensile force in a cord immediately after the rod is given the angular velocity.



Solution: Immediately after the twist, the speeds of points B and D are $\omega L/2$ each and the locus of their path lies on a sphere of radius l. Thus vertical acceleration of both these points must be:



$$a_y = \frac{v^2}{R} = \frac{\omega^2 L^2}{4l}.$$

This must also be the vertical acceleration of all points of the rod ∵ the rod remains horizontal. Thus making LOM equation in y direction, $\Sigma F_y = \Sigma m \cdot a_y$ ∴

$$2(T+\Delta T) - mg = ma_y = \frac{m\omega^2 L^2}{4l}$$

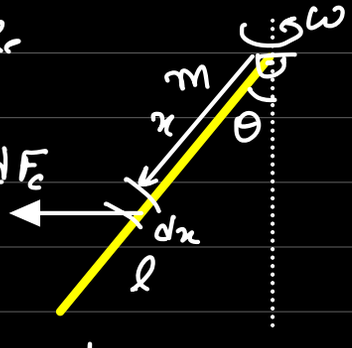
where $T = mg/2$ and ΔT is incremental tension.

Thus

$$\Delta T = \frac{m\omega^2 L^2}{8l} \text{ Ans.}$$

Example : Irodov : 1.270 : A thin uniform rod of length l and mass m rotates uniformly about a vertical axis with angular velocity ω (The upper end is pivoted) Find the angle θ between the vertical and the rod.

Solution: In a frame rotating with ω on the axis, the rod appears stationary, thus, torque of centrifugal forces must be balanced dF_c by the gravitational torque. Now since centrifugal field is a non-uniform field, we need to find its torque by integration of elemental torques.



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Now consider an element of the rod of thickness dx at a distance x from the pivot. Due to centrifugal force, :

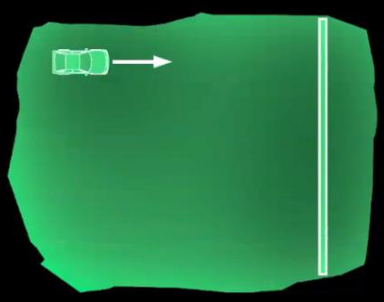
$$d\tau = \left[\frac{m}{l}dx\,\omega^2 x\sin\theta\right] x\cos\theta \qquad (\because F \times r_\perp)$$

$$\Rightarrow \tau = \frac{m\omega^2}{l}\sin\theta\cos\theta\int_0^l x^2 dx = \frac{m\omega^2 l^2}{3}\sin\theta\cos\theta$$

This must be balanced by torque of gravity, $\tau_g = mg\frac{l}{2}\sin\theta$. Thus:

$$mg\frac{l}{2}\sin\theta = \frac{m\omega^2 l^2}{3}\sin\theta\cos\theta \Rightarrow \cos\theta = \frac{3g}{2\omega^2 l} \text{ Ans.}$$

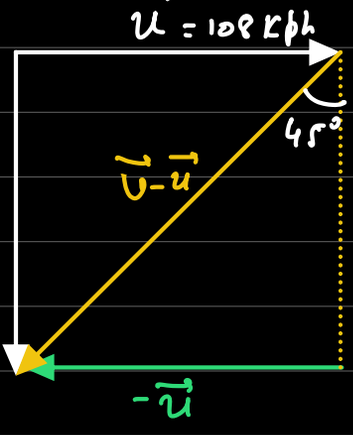
EXAMPLE: PATHFINDER MCQ38:



A car is moving with speed 108 km/h on a large uniform horizontal pavement perpendicularly towards a wall. To avoid collision, the driver takes a turn and finally runs the car parallel to the wall with the same speed. Coefficient of friction between the tyres and the pavement is 0.6, acceleration of free fall is 10 m/s^2 and all the load of the car is concentrated on the front axle. If the turning process is the fastest one on this pavement, check validity of the following statements.

(a) The car follows a circular path during the turn.
(b) The car follows a parabolic path during the turn.
(c) Minimum radius of curvature of the path is 75 m.
(d) Initial distance of the wall must be greater than $75\sqrt{2}$ m.

Solution: For fastest turning acceleration must always be directed along net change in velocity vector. The change in velocity vector is shown in adjacent diagram. Also since we want fastest change in velocity, we must keep maximum possible magnitude of acceleration. Thus net acceleration is μg along the orange vector. Since it is a case of constant acceleration, path is like that of a projectile, i.e a parabolic path, as if it were a projectile launched at an angle 45° with horizontal. Thus min radius of curvature, $R = (u\cos 45°)^2/\mu g$



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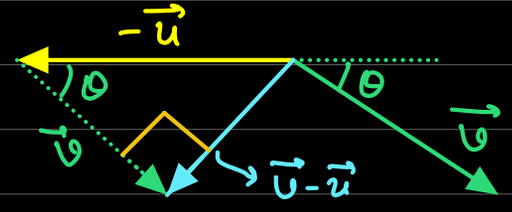
$u = 108 \times \frac{5}{18} = 30\ ms^{-1}$ thus $s_2 = \left(\frac{30}{\sqrt{2}}\right)^2 \times \frac{1}{6} = 75\ m.$

Now component of acceleration $\perp$ to wall, $a_n = \mu g \cos 45° = 6/\sqrt{2}$
Let s be min allowable distance from wall. Now: $v_n^2 - u_n^2 = 2a_n s_n$
thus: $0^2 - 30^2 = -2 \times \frac{6}{\sqrt{2}} \times s \Rightarrow s = 75\sqrt{2}\ m.$

Thus correct options are (b), (c) and (d).

Think: Suppose it were not necessary to have same final speed as initial, then what would be the shortest time to turn through an angle θ given initial speed is u $\perp$ to the wall. (Coefficient of friction is μ).

Solution: The problem essentially reduces to finding minimum possible magnitude of $\Delta\vec{v}$ with $\vec{v}$ being at angle θ to $\vec{u}$.
For this we draw the vector diagram as shown in the figure (draw $\perp$ on line of $\vec{v}$ to find shortest magnitude of $\vec{v} - \vec{u}$.
Accordingly, shortest time, $t_{min} = \frac{u \sin\theta}{\mu g}$.



(For a 90° turn this reduces to the case of final speed becoming zero).

EXAMPLE: PATHFINDER MCQ 40:

A particle is moving with constant angular velocity on a circular path of radius R in the x-y plane. If observed from a reference frame moving with constant velocity along the z-axis, the particle will appear moving on a helical path of constant pitch h. Making use of the given information, what expression can be deduced for radius of curvature of the helical path.

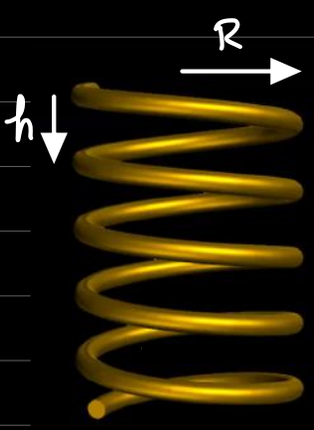
(a) $\sqrt{R^2 + \frac{h^2}{4\pi^2}}$ (b) $R + \frac{h^2}{4\pi^2 R}$

(c) $\sqrt{R^2 + h^2}$

(d) Cannot be calculated from the given information.

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Solution: Let the centripetal acceleration of the particle be a_c and speed in circular motion be u then $R = u^2/a_c$. Now let speed of the observer along z axis be v, then $h = \frac{2\pi}{\omega} \times v = \frac{2\pi R v}{u}$



$\Rightarrow v = uh/2\pi R$

Now from this frame, net speed of particle:

$V = \sqrt{u^2 + v^2}$ and radius of curvature:

$$R_c = V^2/a_c = \frac{u^2 + u^2h^2/4\pi^2R^2}{a_c} = \frac{u^2}{a_c}\left(1 + \frac{h^2}{4\pi^2R^2}\right)$$

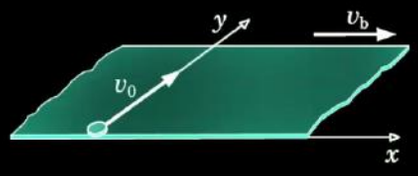
$$= R\left(1 + \frac{h^2}{4\pi^2R^2}\right) = R + \frac{h^2}{4\pi^2R}$$ thus:

$$\boxed{R_c = R + \frac{h^2}{4\pi^2R}}$$ option (b) ans.

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EXAMPLE: PATHFINDER MCQ 42-44: MODIFIED:

A horizontal conveyor belt is running at a constant speed $v_b = 3.0$ m/s. A small disc enters the belt moving horizontally with a velocity $v_0 = 4.0$ m/s that is perpendicular to the velocity of the belt. Coefficient of friction between the disc and the belt is 0.50.



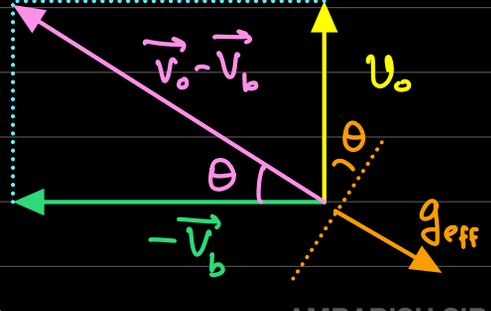
(a) What is the initial radius of curvature as seen from ground

(b) What is the minimum speed as seen from ground

(c) What is the minimum width of conveyor so that disc always remains on it.

Solution: From conveyor frame, the initial velocity of the disc is $\vec{v}_0 - \vec{v}_b$ as shown and friction always acts against this line.

$\therefore$ Path of the disc is like a projectile as seen from ground frame. Thus minimum speed from ground frame is $v_0 \cos\theta$ i.e:



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$$v_{min} = \frac{v_0 v_b}{\sqrt{v_0^2 + v_b^2}} \quad \text{Ans. (b)}$$

Now initial radius of curvature, $R = v_0^2 / g_{eff} \cos\theta = \dfrac{v_0^2 \sqrt{v_b^2 + v_0^2}}{\mu g \times v_b}$

Thus $R = \dfrac{v_0^2 \sqrt{v_b^2 + v_0^2}}{\mu g v_b}$ Ans (a)

Now let y_{max} be the max y coordinate of the disc. Then:

$$0^2 - v_0^2 = -2 g_{eff} \sin\theta \, y_{max} \Rightarrow y_{max} = \frac{v_0^2}{2 g_{eff} \sin\theta} = \frac{v_0^2 \sqrt{v_0^2 + v_b^2}}{2 \mu g v_0}$$

thus $y_{max} = \dfrac{v_0 \sqrt{v_0^2 + v_b^2}}{2 \mu g}$ Ans (c).

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